

1. Administrative & Study Identification

Full Study Title: Estimated Vaccine Effectiveness and Durability of Pfizer/BioNTech 2024-2025 COVID-19 Vaccine **Short Title/ Acronym:** Pfizer/BioNTech 2024-2025 COVID-19 VE Study **Protocol Number:** C4591068 **NCT Number:** NCT06703190 **Sponsor Name:** Pfizer **Investigational Product:** Pfizer-BioNTech BNT162b2 COVID-19 Vaccine (2024-2025 KP.2 formulation) **Phase of Development:** N/A (Retrospective Observational Study) **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the real-world effectiveness of the 2024-2025 Pfizer/BioNTech COVID-19 vaccine formulation against symptomatic SARS-CoV-2 infection in patients \$\\ge\$ 5\$ years of age testing at CVS MinuteClinics. **Secondary Objectives:** To assess vaccine effectiveness (VE) across different age groups, during periods when different subvariants are circulating, and by months since the receipt of the 2024-2025 formulated COVID-19 vaccine dose.

2.2 Background & Rationale Omicron and its sublineages have comprised the majority of sequenced SARS-CoV-2 genomes since December 2021. Throughout 2024, FLiRT-mutated subvariants (including the KP.2 strain) became the predominant variants in the US. These sub-lineages are antigenically distant from prior strains. The 2024-2025 updated KP.2-adapted vaccine generates improved responses against these circulating sublineages compared to previous XBB-adapted formulations. This study aims to provide real-world evidence on the effectiveness of this updated vaccine in preventing symptomatic COVID-19.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Retrospective Test-Negative Case-Control Design) **Control Type:** Unvaccinated (No receipt of the 2024-2025 formulation) **Blinding:** N/A (Open-label / Database analysis) **Randomization:** N/A (Observational)

3.2 Planned Enrollment & Duration Target \$N\$: N/A (Large-scale retrospective database analysis) **Number of Sites:** CVS MinuteClinic locations across the United States **Estimated Study Start:** September 5, 2024 **Estimated Primary Completion:** Terminated (Study was terminated due to incomplete data linkage on KP.2 vaccination status; not due to safety concerns).

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. MinuteClinic Patients (≥ 5 years of age) testing for SARS-CoV-2 on or after September 5th, 2024 (at least 14 days after FDA approval of the updated formulation). 2. At least one COVID-19 symptom reported on the ICATT questionnaire. 3. No evidence of a positive SARS-CoV-2 test, COVID-19 diagnosis, or COVID-19 antiviral medications picked up at CVS within the prior 90 days.

4.2 Exclusion Criteria 1. Patients testing at drive-thru locations or without a completed ICATT questionnaire; patients with invalid test results or not reporting any symptoms. 2. Patients who received a 2024-2025 formulated COVID-19 vaccine other than from Pfizer/BioNTech. 3. Patients who received a 2024-2025 formulated COVID-19 vaccine < 14 days prior to the test date.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose of the 2024-2025 formulated Pfizer-BioNTech BNT162b2 vaccine. **Route of Administration:** Intramuscular (IM) injection. **Duration of Treatment:** Single administration; effectiveness evaluated ≥ 14 days post-vaccination.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications. **Prohibited:** Receipt of alternative (non-Pfizer/BioNTech) 2024-2025 COVID-19 vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-14 to -7	CVS MinuteClinic Visit: SARS-CoV-2 rapid antigen test (RAT) and completion of ICATT symptom questionnaire.
2	0 to 14	Vaccination Assessment
3	14 to 28	Retrospective validation of vaccination status via CVS or Aetna records ≥ 14 days prior to the test.

(Note: As a retrospective database study, there were no prospective clinical visits or active patient follow-ups).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Estimated Vaccine Effectiveness (VE) against symptomatic SARS-CoV-2 infection. Calculated as $1 - \text{odds ratio}$ from logistic regression models multiplied by 100. **Measurement Method:** Cases defined as patients testing

positive for SARS-CoV-2 by RAT reporting ≥ 1 symptom. Controls defined as testing negative by RAT reporting ≥ 1 symptom.

7.2 Secondary & Exploratory Endpoints [Endpoint 1] VE across different age groups. [Endpoint 2] VE by months since receipt of the 2024-2025 formulated vaccine and across periods of different circulating subvariants.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: N/A (This is a retrospective secondary data analysis of effectiveness, not a prospective safety trial). **Serious Adverse Events (SAEs):** N/A. (Study termination was explicitly noted as not being based on any safety concerns or regulatory requirements). **Data Monitoring Committee (DMC):** N/A (Internal study oversight provided by Pfizer and CVS Health).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Symptomatic CVS MinuteClinic patients with valid SARS-CoV-2 RAT results and known vaccination status. **Sample Size Justification:** Based on available test-negative cases and controls visiting CVS locations from September to December 2024. **Handling of Missing Data:** The study was ultimately terminated/withdrawn from preprint because it was discovered that a data source with vaccination information was incorrectly linked, resulting in individuals being inappropriately dropped due to missing vaccine information.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with ethical guidelines for retrospective observational research; ethical approval was waived by Sterling IRB. **Monitoring Plan:** Retrospective data quality checks and validation against CVS/Aetna vaccination registries. **Audit Trail:** Study data sets underwent quality validation, which ultimately identified the data linkage error leading to the withdrawal/termination of the study.

11. Study Identifiers & Governance

Primary ID: C4591068 **Registry Number:** NCT06703190 **Sponsor/Lead Organization:** Pfizer **Study Title:** Estimated Vaccine Effectiveness and Durability of Pfizer/BioNTech 2024-2025 COVID-19 Vaccine

12. Clinical Framework (Hyperkalemia Specific) (Adapted for Trial Scope)

Primary Condition: Symptomatic COVID-19 (SARS-CoV-2 Infection)
Diagnosis Threshold: Positive SARS-CoV-2 Rapid Antigen Test with
\$\\ge 1\$ symptom **Medical Methodology:** Test-negative case-control
study design **Optimization Target:** Vaccine Effectiveness (VE)
modeling via logistic regression

13. Study Procedures & Methodology

Management Pathway: CVS MinuteClinic SARS-CoV-2 testing and
evaluation pathway **Education Protocol (HCP):** N/A (Retrospective
observational study) **Education Protocol (Patient):** N/A
(Retrospective observational study) **Quality Audit Mechanism:** Data
validation and exposure linkage against CVS or Aetna vaccination
records

14. Observation & Follow-Up Schedule

Total Duration: Retrospective evaluation timeframe (September -
December 2024) **Onsite Visit Cadence:** N/A (Database study) **Remote
Monitoring Frequency:** N/A (Database study) **Checklist Review
Interval:** N/A (Database study)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	Pfizer Medical Lead	____	02- Apr-2026	Clinical Operations Lead	Pfizer/CVS Rep	____	
02-Apr-2026	Lead Statistician	Pfizer Biostatistics	____				
02-Apr-2026							